Function Detective

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Game Concept

Game Master

Your role:
You (the AI) will propose inputs for a hidden function so you can infer it.

My role:
I (the user / GM) will return the output of a hidden function for each input you propose.

Goal:
Infer the exact function:
$**SIGNATURE_OF_CURRENT_FUNCTION**$

Context:
Category: $CATEGORY$.
This function takes NUM_PARAMS parameter(s) with names and types:
$PARAM_LIST$
(Format shown below.)
.
.
.

Output: 0

Player 1

(x: int, y: int) -> int

Context:
Category: math.
This function takes 2 parameter(s) with names and types:
x: int, y: int
(Format shown below.)

Input: 1,1

Input: 2,-3

Experiment design considering the RQ

- Research question:
- Is the performance of a LLM model
  - a function of problem hardness or
  - do different LLM models react differently to the range of problems.
- The outcome we want, 2 types of evaluations:
  - Evaluate performance on **hardness**
  - Evaluate performance on **range of problems**

The 3x5 Experiment Grid

Difficulty

1. Easy
2. Medium
3. Hard

Range (Have several domains)

1. Scalar math (1 input)
2. Pair math (2 inputs)
3. List sequence
4. String
5. Logic (logic/ true false)

What this grid reveals

- Hardness effect:
  - Easy, Medium, Hard
  - If the performance drops across hardness, then that answers, hardness reduces the model's performance
- Range Effect:
  - At a fixed difficulty level, compare the results across domains
  - If the model is much better on some domains than others, we can conclude that "range matters”

Hint mode addition

- A hint mode of the game was introduced where a “hint” is revealed to the model, along with the initial prompt.

Context placeholders:
- Category: `SCALAR_MATH`
- Params: `1`
- Param list: `x: int`

Begin: produce your first `INPUT:` line.

HINT: We are looking for a simple one-variable numeric rule (single input → single output). Consider sign-sensitive behavior.

CANDIDATE FUNCTION IDS (use only these in hypotheses):
- abs_value
- clamp_0_9
- piecewise_linear_break0
- piecewise_linear_break1
- square

On EVERY turn, output one of the following formats.

```python
@register(
    category=Category.SCALAR_MATH,
    difficulty=Difficulty.EASY,
    confusers=["square",
     "piecewise_linear_break0",
     "piecewise_linear_break1"],
    family_id="scalar_basic",
    hint="We are looking for a simple
     one-variable numeric rule (single
     input → single output). Consider
     sign-sensitive behavior."
)
def abs_value(x: int) → int:
    """Return absolute value of x."""
    return abs(x)
```

No penalty mode

- Code is generated on each turn
- This code:
  - If correct → Game ends with win
  - If wrong → Turn gets consumed, game keeps going
- What it aims to measure:
  - Turns till solution
  - Whether errors decrease with turns
  - Do the models use probing or now there is a shift towards brute-guessing

Updates and Changes

- Prompt updated to aim to be more LLM friendly
- Main score = Success × Speed
- $\mathbf{Score} = \boldsymbol{Success} \times \boldsymbol{Speed}$
- where
- $\mathbf{Success} = \begin{cases} 1 & \textit{if all tests pass} \\ 0 & \textit{otherwise} \end{cases}$
- and
- $\mathbf{Speed} = 1 - \frac{\textit{probes used} - 1}{\textit{max probes} - 1}$

Conclusion and future work

- All hypotheses, candidate_ID and confusers have been removed to simplify the game, for now.
- Future work focuses on finalizing a prompt that is:
  - Even simpler
  - Short